

# Lab 1: Cloud Virtualization and Data Center Architecture Report

This report summarizes Lab 1, which explores virtualization technologies, cloud infrastructure, and tail latency behavior using practical experiments with VMs, containers, and Flask applications.

## Objectives

Understand VMs vs Containers, deploy instances, analyze cloud architecture, and study tail latency behavior.

## Tools Used

Docker, Multipass, AWS EC2 (optional), Python Flask, terminal tools (top, htop, ab).

## Part A: VMs vs Containers

Virtual Machines were launched using Multipass while Docker containers were used for lightweight environments. Observations show that containers consume less memory and start faster than VMs.

## Part B: Cloud Infrastructure

AWS EC2 instance was used to explore Nitro Hypervisor. Nitro appears at system level but is not directly accessible, ensuring security isolation. Process monitoring was done using top/htop.

## Part C: Tail Latency

A Flask application with randomized delay was tested using Apache Benchmark. Results show a skewed distribution where most requests are fast but a few exhibit high latency (tail latency effect).

## Conclusion

Containers are more efficient for microservices due to lightweight design, while VMs provide stronger isolation. Tail latency is an important factor in distributed systems performance.